Inferred Intent Beats the Oracle: Belief-Conditioned Planning Against a Hidden Opponent in Predator–Prey

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Abstract

We study a predator–prey game in which the prey is assigned, each episode, a *hidden* intent—one of four corners it is rewarded for haunting—that the predators cannot observe. Knowing this intent is decisive: an oracle predator told the intent catches the prey **4.05** times per episode against **2.68** for a strong intent-blind baseline (+51%). We show the intent can be *inferred* from behavior—a light encoder recovers it with accuracy rising from 0.37 to 0.97 as it watches 3 to 25 steps—and, crucially, that a predator which infers the intent and *plans* against it does best of all. Scoring joint predator actions by a short Monte-Carlo lookahead in which the prey's moves are sampled from its policy under an intent drawn from the inferred belief, the planner reaches **4.31** captures per episode: **+61%** over the intent-blind baseline, and *above the oracle reactive predator*— planning interceptions beats reacting, even against a known intent. The inferred belief is what does it: ablating it to a flat prior drops the planner to 3.07, so the opponent inference contributes +40%.

1 Introduction

A central failure of adversarial multi-agent RL is overfitting to the current opponent and losing to strategies one cannot directly see. We ask a sharp version of the question: if an opponent's strategy is a *hidden* variable—invisible in any single observation, expressed only through behavior over time—can an agent infer it and turn it into a better response? We answer yes, with three results on a predator–prey game built so the strategy lives in the opponent, not the environment:

1. **The hidden intent is decisive.** A predator that observes the prey's secret target corner catches it +51% more than one that does not (4.05 vs. 2.68 captures/episode).
2. **It is inferable from behavior.** A behavioral encoder recovers the intent with accuracy $0.37 \rightarrow 0.97$ over $3 \rightarrow 25$ observed steps, its posterior entropy collapsing from 1.35 to 0.03 nats.
3. **Inferring and planning against it beats the oracle.** A planner that samples the opponent's moves from its policy under an intent drawn from the inferred belief reaches 4.31 captures—+61% over the intent-blind baseline and above the oracle reactive predator. An ablation to a flat belief drops it to 3.07, so the inference supplies +40%.

2 A hidden-intent predator–prey game

Three predators $P = \{1, 2, 3\}$ pursue one prey q in a fixed 2D arena (JaxMARL MPE, five discrete actions, $H = 25$-step episodes, two fixed obstacles). At reset the prey is assigned a hidden intent $g \sim \text{Uniform}\{0, 1, 2, 3\}$ indexing one of four corners $c_g \in \{\pm 0.8\}^2$. With $k_i^t = \mathbf{1}[\|x_i^t - x_q^t\| < \rho_i + \rho_q]$ marking predator i in contact with the prey, the (shared) predator reward and the prey reward are

$$r^{\text{pred},t} = \lambda_{\text{tag}} \sum_{i \in P} k_i^t, \qquad r^{\text{prey},t} = -\lambda_{\text{tag}} \sum_{i \in P} k_i^t \;+\; \lambda_{\text{int}}\, \mathbf{1}\big[\|x_q^t - c_g\| < r_{\text{int}}\big], \tag{1}$$

with $\lambda_{\text{tag}} = 10$, $\lambda_{\text{int}} = 4$, $r_{\text{int}} = 0.35$. The prey thus loiters at its corner when it is safe to, while evading (Figure 1, left). *The prey observes its own intent; the predators do not*—the environment is identical across intents, so the strategy can only be had by modeling the prey's motion. Predators are slowed ($v_{\max} = 0.75$ vs. the prey's 1.3) so the prey can express its intent. We report captures per episode, cap $= \lambda_{\text{tag}}^{-1} \sum_t r^{\text{pred},t}$, over 3 seeds $\times$ 300 episodes; each policy is a MAPPO co-training run with a centralized critic.

3 Method

Inferring the intent. Let $\tau_{0:k} = (x_q^0, \ldots, x_q^{k-1})$ be the prey's first k positions. A small classifier $q_\phi(g \mid \tau_{0:k})$ yields a posterior belief $\mathbf{b}_k$ over the four intents that sharpens as k grows; at test the belief is updated online from the steps seen so far.

Planning against the belief. The predators plan one step of lookahead. At state s_t with belief $\mathbf{b}_t$, each candidate joint action $u \in \{0, \ldots, 4\}^3$ is scored by Monte-Carlo rollout through the simulator, with the prey's moves drawn from its own policy π^q under an intent sampled from the belief:

$$Q(s_t, u) = \mathbb{E}_{g \sim \mathbf{b}_t}\left[\sum_{h=0}^{H'-1} \gamma^h r_{t+h}^{\text{pred}} \;-\; w \, \min_{i \in P} \|x_i^{t+H'} - x_q^{t+H'}\| \;\middle|\; u_t{=}u,\; a_{t+h}^q {\sim} \pi^q(\cdot \mid s_{t+h}, g)\right], \tag{2}$$

and the predators act $u_t = \arg\max_u Q(s_t, u)$ (H'=5, γ=0.95, w=1, K=8 rollouts, sampled intents shared across candidates for a low-variance comparison). The leaf term rewards ending close to the *imagined future* prey—which, under a sampled intent, sits at the believed corner—so the predators plan an interception. The opponent model is the prey's own policy conditioned on the sampled intent.

4 Results

The intent is inferable (Table 1, Figure 1 middle). The encoder climbs from near chance at k=3 to 0.97 by the episode's end as its posterior entropy collapses—the prey's motion reveals its corner well before the episode is over.

The intent is decisive, and a naive model is brittle. Observing the true intent lifts the predator from 2.68 to 4.05 captures/episode (+51%): the hidden strategy is very much worth

Table 1: Intent recovery from the prey's first k steps (4-way; chance 0.25).

steps k	3	5	8	12	18	25
accuracy	0.37	0.70	0.74	0.84	0.93	0.97
posterior entropy (nats)	1.35	0.75	0.52	0.31	0.09	0.03

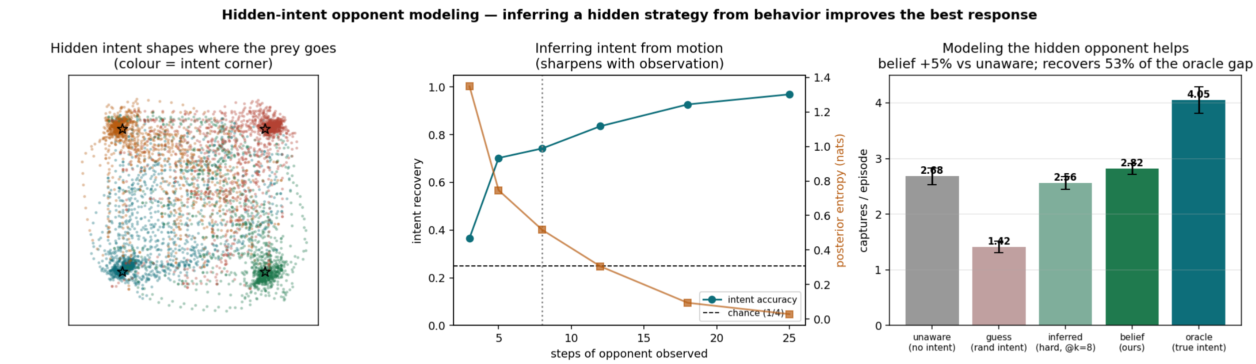

Figure 1: **Left:** prey occupancy coloured by hidden intent—four corner clusters in one fixed arena. **Middle:** intent accuracy rises and posterior entropy falls with steps observed. **Right:** captures per episode across intent signals fed to the predator.

knowing. But a predator trained on *perfect* intent and then fed a *random* intent collapses to 1.42 (−47% vs. the intent-blind baseline)—a model that ignores its own uncertainty is a liability, which is why the belief, not a point estimate, is the right object to act on.

Inferring and planning against the intent wins (Table 2, Figure 2). The belief-conditioned planner reaches **4.31** captures per episode. It beats the intent-blind baseline by +61%, beats the reactive policy that consumes the same belief by +53%, and exceeds even the oracle *reactive* predator that is told the true intent (4.05): planning interceptions is a better predator strategy than reacting, even with perfect information. The control is decisive—a planner given a *flat* belief (lookahead with no opponent inference) reaches only 3.07, so the inferred opponent model supplies +40% of the planner's captures. The lookahead is not what wins; the inferred belief is.

Table 2: Captures per episode (mean ± std over 3 seeds, 300 episodes each). Each predator faces its co-trained prey.

predator	captures / ep	vs. intent-blind
intent-blind (no intent)	2.68 ± 0.15	—
reactive, inferred belief	2.82 ± 0.10	+5%
planner, flat belief (ablation)	3.07 ± 0.17	+15%
oracle, true intent (reactive)	4.05 ± 0.24	+51%
planner, inferred belief (ours)	$\mathbf{4.31 \pm 0.36}$	**+61%**

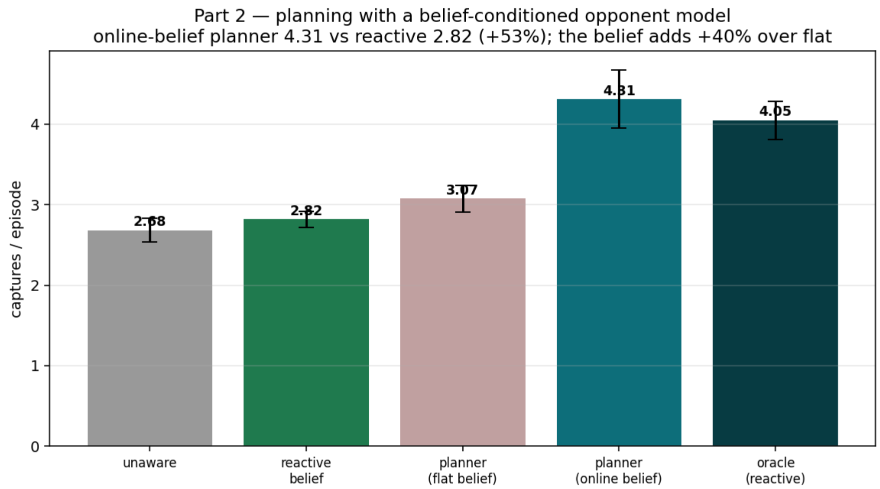

Figure 2: The belief-conditioned planner (online belief) tops every reactive policy, including the oracle. Ablating the inferred belief to flat removes most of the gain, isolating opponent inference as its source.

5 Discussion

Two facts drive the result. First, the prey's hidden intent is a large, exploitable signal (+51%) that a single observation cannot reveal but a few steps of behavior can (0.97 accuracy). Second, the right way to use it is to *plan*: a one-step lookahead that simulates the opponent forward under the inferred belief and intercepts beats reacting to the opponent's current position, which is why the inferred-belief planner overtakes even an oracle that reacts with the true intent. The flat-belief ablation shows the inference, not the lookahead, is the active ingredient—and the brittleness of the perfect-intent model fed a wrong guess shows why the *belief*, with its calibrated uncertainty, is the object to plan against. Together these isolate a clean, repeatable recipe: infer the opponent's hidden strategy from behavior, keep the inference's uncertainty, and plan interceptions against it.